National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Engineering Management	Course Code:	MG330
Program:	BS(Electrical Engineering)	Semester:	Fall 2017
Duration:	150 Minutes	Total Marks:	50
Paper Date:	20-December-17	Weight	50%
Section:	ALL	Page(s):	2
Exam Type:	Final		

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Instruction/Notes: *Start each question from new page, show all steps, and assume if something is missing in paper.*

Q #1: Draw a PERT diagram and find the critical path (10 marks)

Activity	Activity Description	Immediate Predecessors	Estimated Duration
A	Excavate	-	2 Weeks
B	Lay the foundation	A	4 Weeks
C	Put up the rough wall	B	10 Weeks
D	Put up the roof	C	6 Weeks
E	Install the exterior plumbing	C	4 Weeks
F	Install the interior plumbing	E	5 Weeks
G	Put up the exterior siding	D	7 Weeks
H	Put up the exterior painting	E,G	9 Weeks
I	Do the electrical work	C	7 Weeks
J	Put up the wall board	F,I	8 Weeks
K	Install the flooring	J	4 Weeks
L	Do the interior painting	J	5 Weeks
M	Install the exterior fixtures	H	2 Weeks
N	Install the interior fixtures	K,L	6 Weeks

Q #2: Define the objective of forecasting, what are the steps by which we can implement forecasting.

(5 marks)

Q #3: Describe the five possibilities which range from no structure to a totally separate project organization.

(10 marks)

2 Q#4: According to Pinto what are the most important influences promoting a project management? (5 marks)

Q#5: Name the five life-cycle phases to classify Costs.

(5 marks)

4 Q#6: Describe the steps to be carried out to develop LCC model once the agreement is reached on the types of analyses. CBS

Cost estimating

(5 marks)

Q#7: One of four new techniques, or the current method, can be used to control mildly irritating chemical fume leakage into the surrounding air from a mixing machine. The estimated costs and benefits (in the form of reduced employee health costs) are given below for each method. Assuming that all methods have a 10-year life with zero salvage value, determine which one should be selected, using a MARR of 15% per year and the incremental B/C method. (10 marks)

	Technique			
	1	2	3	4
Installed cost, $	15,000	19,000	25,000	33,000
AOC, $/year	10,000	12,000	9,000	11,000
Benefits, $/year	15,000	20,000	19,000	22,000

and the incremental B/C method

	Technique			
	1	2	3	4
Installed cost, $	15,000	19,000	25,000	33,000
AOC, $/year	10,000	12,000	9,000	11,000
Benefits, $/year	15,000	20,000	19,000	22,000

Formulas

$\left(\frac{F}{P}, i, n\right) => F = P(1+i)^n$	$\left(\frac{P}{F}, i, n\right) => P = \frac{F}{(1+i)^n}$
$\left(\frac{P}{A}, i, n\right) => P = A\frac{(1+i)^n - 1}{i(1+i)^n}$	$\left(\frac{A}{P}, i, n\right) => A = P\frac{i(1+i)^n}{(1+i)^n - 1}$
$\left(\frac{F}{A}, i, n\right) => F = A\frac{(1+i)^n - 1}{i}$	$\left(\frac{A}{F}, i, n\right) => A = F\frac{i}{(1+i)^n - 1}$